

Seminar on Gestalten

Regensburg - SoSe 2026

February 2, 2026

In this seminar we will discuss Scholze's lecture course [Sc26] on Gestalten.

Logistics: We will meet weekly on Monday 12(ct)-14 in M311.

Talks:

1	13/04/26	Presentable ∞-categories: Source: [Sc26, Lecture II] and [Sc26, Lecture III]
2	20/04/26	Quasicoherent sheaves of higher categories I: Source: [Sc26, Lecture IV] up to §4.3
3	27/04/26	Descent for quasicoherent sheaves: Source: [Sc26, Lecture IV] from §4.3
4	04/05/26	Quasicoherent sheaves of higher categories II: Source: [Sc26, Lecture V]
5	11/05/26	Stefanich Rings: Source: [Sc26, Lecture VI] up to §6.4
6	18/05/26	Proper and étale maps of Stefanich rings: Source: [Sc26, Lecture VI] from §6.4
7	01/06/26	Gestalten: Source: [Sc26, Lecture VII]
8	08/06/26	Examples: Source: [Sc26, Lecture VIII]
9	15/06/26	Descendability: Source: [Sc26, Appendix to Lecture VIII]
10	22/06/26	Six functors and Gestalten: Source: [Sc26, Lecture IX], up to extension to stacks
11	29/06/26	The Gestalt $[\mathrm{Spec}(\mathbb{Z})]_{\mathrm{SH}}$: Source: [Sc26, Lecture IX + Appendix]
12	06/07/26	Cartier duality: Source: [Sc26, Lecture X]
13	13/07/26	The proof of Cartier duality: Sketch the proof of Cartier duality. Source: [Sc26, Lecture XI] and [Sc26, Lecture XII]

References

[Sc26] Peter Scholze, *Geometry and Higher Category Theory*. 2026. available at the author's website.